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REVIEW

# Pyrolysis: a promising technology for agricultural waste conversion into value-added products

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**Abstract**
Agricultural waste management poses a significant challenge due to increasing generation rate of this residue, necessitating the development of solutions to minimize various problems in this context. Among the potential possibilities, pyrolysis stands out as a promising technology due to its low operational costs, simplicity of operation, and the production of several value-added products, including solids (biochar), liquids (pyroligneous extracts: pyroligneous acid and bio-oil), and gaseous (non-condensable gases). This work aimed to synthesize the recent advances and potential uses for the products obtained from the pyrolysis of agricultural wastes, with a particular focus on the emerging applications of the liquid fraction. To achieve the purpose of this review, a bibliographic portfolio was created using a state-of-the-art approach, utilizing Scopus as the database. The yields and uses of pyrolysis products vary according to operational parameters, such as temperature, heating rate, residence time, reactor, agro-waste characteristics, and others. These products have diverse potential applications, including environmental uses (soil remediation, wastewater treatment, and bioenergy), pharmaceutical purposes (antimicrobials, antioxidants, anti-inflammatories), food industry uses (preservatives and smoke flavor), and applications in agriculture and livestock (plant growth promotion, pest and disease control, feed additive, and soil conditioner). Therefore, pyrolysis is a promising technology for the valorization of agricultural wastes, promoting various aspects of the circular economy and providing a sustainable solution to obtain several value-added products with diverse applications from this thermochemical conversion process.

**Keywords** Biochar · Bio-oil · Wood vinegar · Pyroligneous acid · Syngas · Waste valorization

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## 1 Introduction

The increase in waste generation due to population growth, combined with the complexity of its management, poses challenges for several nations worldwide (Luttenberger, 2020; Prajapati et al., 2021). Much of the waste is still poorly managed, with low indicators for collection, segregation, and disposal, resulting in substantial amounts being either burned in the open sites or deposited in unsanitary landfills, particularly in underdeveloped or developing countries (Prajapati et al., 2021; Xu et al., 2011). This presents a growing environmental issue, as inadequate waste management practices lead to various negative impacts on the environment and human health (Khan et al., 2022; Prajapati et al., 2021).

Among the different types of waste, the generation of agricultural waste continues to increase due to the demand for food, particularly in countries with high agricultural production (Freitas et al., 2021). The rise in agricultural production and, consequently, the increase in agricultural waste generation, have spurred the development of technologies and strategies to contribute to the more sustainable management of such waste (Gavilanes-Terán et al., 2015; Zainudin et al., 2022). As an alternative to the conventional disposal of these residues, the scientific community has reported numerous potential uses and conversion routes for agricultural wastes, including: (a) production of biofertilizers through composting (Cortés et al., 2020; Ferrari et al., 2019; Singh et al., 2022; Zainudin et al., 2022); (b) biofuels (e.g. bioethanol, biomethane, biodiesel, biohydrogen) (Colla et al., 2019; Ferreira et al., 2021; Gopinath et al., 2016; Singh et al., 2022); (c) animal feed (Chilakamarry et al., 2022; Koul et al., 2022); (d) microbial biomass growth (Nazari et al., 2021); (e) enzymes production (Chilakamarry et al., 2022; Kreling et al., 2020; Nazari et al., 2022); (f) bioactive compounds production (Santos et al., 2022; Singh et al., 2022); (g) production of organic acids (e.g. citric, lactic and glutamic acid) (Chilakamarry et al., 2022; Gopinath et al., 2016); and others.

However, for these wastes to become valuable feedstocks for diverse economic segments, it is necessary to develop technological solutions capable of adding value to these materials (Ferrari et al., 2019; Xu et al., 2009, 2011). The implementation of technologies that convert wastes into useful materials promotes the circular economy, resulting in greater sustainability throughout the production chain: from waste generation to its reinsertion as raw material in a different process (Luttenberger, 2020). One way to achieve this goal is through pyrolysis, a thermochemical process carried out in low or oxygen-free atmospheres, which converts wastes into three distinct products: solid, liquid, and gases, all with potential applications in several areas (Freitas et al., 2021; Matos et al., 2021; Xu et al., 2011). Among these products, most of the research has focused on the potential of biochar for environmental and agricultural uses, and bio-oil as a biofuel (Ben et al., 2015; Mullen et al., 2010). However, the development of value-added products from the pyrolysis of agricultural wastes is a promising area that warrants further investigation.

Several reviews focusing on the pyrolysis of agricultural residues and its products have been published in recent years, covering various aspects such as (a) pyrolysis for the production of biofuels (Douvartzides et al., 2022; Ge et al., 2021), formic acid (Shen et al., 2021), biosilica and bioenergy (Nawaz et al. 2022), bio-oils for fuel and chemicals (Mohammed et al., 2023); (b) production and/or application of biochar (Kulkarni et al., 2023; Shen et al., 2024; Tan, 2023; Tan et al., 2023; Yrjälä et al., 2022); (c) life cycle assessment of pyrolysis or its products (James et al., 2022; Yu et al., 2022; Zhu et al., 2022); (d) current and

future trends in biomass pyrolysis (Fahmy et al., 2020); among others. A similar review has been previously published (Fodah et al., 2022). However, these authors specifically focused on the production and applications of biochar, bio-oil, and gas from microwave-assisted pyrolysis of agro-residues. To the best of our knowledge, the present review distinguishes itself from the others by compiling recent advances in obtaining solid, liquid, and gaseous products from the pyrolysis of agricultural wastes, emphasizing their various potential applications, with a particular focus on the use of the liquid fraction (pyroligneous extract) in agriculture, medicine, and the food industry.

## 2 Methodology

This review employed a state-of-the-art approach, utilizing the Scopus database (scopus.com) to create a bibliographic portfolio. The works were selected by inserting the following terms in the database search: "pyrolysis" AND ("agricultural waste" OR "agricultural residue" OR "agro-waste"). Some delimitations were applied during the selection process: (a) timespan: 2010–2023; (b) search within: article title and abstract; (c) document type: article and review; (d) language: English; (e) source type: journal. In the initial search, 725 articles and 81 reviews were identified, totaling 806 documents. After a content analysis (title, abstract and objective), the main bibliographic portfolio was composed of 113 papers. Figure 1 presents the methodological flowchart of this work.

## 3 Pyrolysis of agricultural wastes

Thermochemical conversion processes emerge as a promising approach to efficiently and economically transform biomass and waste, generating products for various applications. In comparison to biochemical processes, they offer advantages such as shorter reaction times, applicability to all types of biomasses, high conversion efficiency, and a variety of products (Okolie et al., 2022; Perera et al., 2021). Gasification, liquefaction, combustion, torrefaction and pyrolysis are examples of thermochemical conversion processes, and their characteristics, advantages, and disadvantages are detailed in Table 1.

Among the thermochemical conversion technologies, pyrolysis is a potential alternative due to its profitability, cost-effectiveness, and ease of operation. This process involves the thermal degradation of biomass components in an inert atmosphere or under limited oxygen

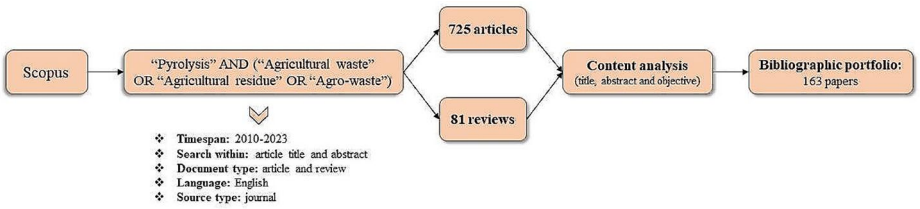



**Fig. 1** Methodological flowchart for creating the bibliographic portfolio. Query: Database: Scopus; Terms of search: "Pyrolysis" AND ("Agricultural waste" OR "Agricultural residue" OR "Agro-waste"); Search within: title and abstract; Timespan: 2010–2023; Document type: article and review; Language: English; Source type: journal

**Table 1** Comparison of thermochemical processes: characteristics, advantages, and disadvantages (Das et al., 2021; Okolie et al., 2022; Rahimi et al., 2022; Wang & Wu, 2023)

| Technologies | Characteristics | Advantages | Disadvantages |
|---|---|---|---|
| Gasification | Thermochemical conversion to gas in the presence of a gasification agent; can be conventional (air, oxygen, steam; 500–1400 °C) or hydrothermal (supercritical water; pressure > 21 MPa) | • Higher gas yield (mainly $H_2$)<br>• Raw material with high moisture content (hydrothermal)<br>• Energy efficiency | • Complex process<br>• High implementation, operational, and maintenance costs<br>• Tar and fly ash production<br>• Catalyst deactivation due to coke deposition<br>• Not suitable for raw materials with high moisture content (conventional) |
| Liquefaction | Thermochemical transformation in the presence of a solvent (water or alcohols) at temperatures (200–600 °C) and high pressure (5–25 MPa). Produces bio-oil, char, and gases. | • Allows raw materials with high moisture content<br>• Higher bio-oil yield<br>• Bio-oil with higher heating value and lower oxygen content than pyrolysis | • Complex process<br>• High implementation, operational, and maintenance costs<br>• Specific conditions<br>• Difficult separation, requiring additional processes |
| Combustion | Thermal transformation in the presence of oxygen, generating heat, water, and $CO_2$. | • Simple and practical<br>• Cost-effective<br>• Any type of biomass<br>• Direct heat generation | • Low efficiency<br>• Significant $CO_2$ emission |
| Torrefaction | Thermochemical pretreatment at mild temperatures (200 to 300 °C) and inert atmosphere | • Lower energy expenditure<br>• Ease of operation<br>• Solid with higher energy density and lower moisture for subsequent application | • Biomass needs to be more reactive<br>• Residence time<br>• Operational cost<br>• Ash-related issues |
| Pyrolysis | Thermal decomposition in the absence of oxygen at temperatures of 350–800 ºC, producing bio-oil, char, and gas. The process can be slow, fast, microwave-assisted, or catalytic. | • Operational simplicity<br>• Efficient<br>• Cost-effective<br>• Control over product yield and properties<br>• Energy efficiency (microwave)<br>• Possibility of co-pyrolysis | • Residence time (slow pyrolysis)<br>• Not suitable for raw materials with high moisture content<br>• Catalyst deactivation due to coke deposition (catalytic) |

conditions, yielding products in solid, liquid, and gaseous states (Demiral & Ayan, 2011; Guedes et al., 2018; Matos et al., 2021; Rahimi et al., 2022). Pyrolysis is a distinctive process based on its ability to control conditions, aiming to optimize the yield and properties of desired products. Therefore, pyrolysis is a promising solution for transforming agricultural residues and by-products into value-added products. By enhancing the efficiency of agricultural waste management, pyrolysis plays a significant role in promoting responsible environmental practices and facilitating sustainable economic innovations.

The yields and composition of pyrolysis products depend on the processing conditions (see Table 2). Some parameters strongly influence the yield and composition of pyrolysis products due to their impact on the kinetics of chemical reactions, heat transfer, and reaction time. In addition to the composition of the input biomass, parameters such as particle size, heating and feeding rate, and temperature influence pyrolysis product composition and yields. Pyrolysis can be categorized into slow, intermediate, fast, or flash pyrolysis based

on its operational conditions, including temperature, heating rate, and residence time (Dai et al., 2020; Das et al., 2021; Guedes et al., 2018; Perera et al., 2021; Rahimi et al., 2022).

In slow pyrolysis, the process occurs at slow heating rates (around 0.1-2 °C/s), low to intermediate temperatures (approximately 400 °C), and long solid residence times, which can extend to hours or days (Bridgwater, 2012; Das et al., 2021; Dhyani & Bhaskar, 2018; Encinar et al., 1998; Wu et al., 2022). Slow pyrolysis tolerates more moisture in the raw material and results in high yields of biochar (Das et al., 2021; Encinar et al., 1998; Silos-Llamas et al., 2023). While commonly used for biochar production, liquids can also be obtained from this type of pyrolysis (Dhyani & Bhaskar, 2018; Rahimi et al., 2022). In intermediate pyrolysis, the reaction time is only a few seconds (0.5–10 s) (Bridgwater, 2012), with heating rate between 10 and 200 °C/s (Wu et al., 2022). The process is carried out in temperatures around 300–500 °C and, due to the rapid heating, the particle size must be small (Bridgwater, 2012; Dhyani & Bhaskar, 2018). In fast pyrolysis, the feed is heated at temperatures between 450 and 600 °C, with heating rates around 1000 °C/s and short vapor residence times (<2 s). This type of pyrolysis is associated with a reduction in the yield of solids, and depending on temperature and residence time, either gases or liquids are the highest yielding product (Das et al., 2021; Dhyani & Bhaskar, 2018; Encinar et al., 1998; Perera et al., 2021; Rahimi et al., 2022; Silos-Llamas et al., 2023). Table 2 shows different types of pyrolysis of various agricultural wastes, as well as the reactor types, operational conditions, and the respective yields of products obtained.

Table 2 shows that several agricultural wastes have been explored as potential feedstocks for pyrolysis. Establishing a standardized pyrolysis process is challenging due to the diversity of feedstocks and process parameters, leading to a wide range of products with varying properties. However, slow pyrolysis stands out among the types, particularly for producing biochar and pyroligneous acid. This fact may be linked to the type of reactor used, as reactors coupled to electrically heated ovens are prevalent in most research on biomass pyrolysis. The use of electrical ovens or muffle furnaces is often associated with laboratory equipment retrofitted with fixed bed reactors. These simple systems are unsuitable for performing intermediate or fast pyrolysis, which require more complex equipment like auger or fluidized bed reactors. Additionally, most studies conducted pyrolysis at a process temperature in the range of 500–600 °C and a solids residence time between 60 and 120 min. Solid residence times refer to the duration during which solid biomass or biochar remains inside the reactor, either from the beginning of the heating ramp in batch reactors or from the moment the biomass is inserted into the reactor in continuous reactors. This parameter is directly related to the heating rate of solids, as well as reactor volumes and capacity, it is commonly disclosed for fixed bed and auger systems, where it can be easily manipulated.

The heating of biomass in pyrolysis results in the production of organic vapors, comprising thermally cracked fragments of cellulose, hemicellulose and lignin polymers (Dhyani & Bhaskar, 2018; Hu et al., 2023). These vapors can be condensed to form an organic liquid, commonly known as bio-oil. Non-condensable gases exit the condensing system and can be utilized for bioenergy purposes (Cai et al., 2024). The remaining fraction is biochar, a carbon-rich solid material (Dhyani & Bhaskar, 2018). Temperatures between 390 and 650 °C and shorter vapor residence times enhance liquid yield. On the other hand, at high temperatures (>650 °C) and vapor residence times of over 1 s, secondary cracking of volatiles occurs, leading to a higher gas yield. Lower temperatures prevent the complete decomposition of biomass, increasing the yield of the solid fraction (Encinar et al., 1998;

**Table 2** Yield of solid, liquid and gaseous products obtained from the pyrolysis of various agricultural wastes in different types of reactors and experimental conditions

| Biomass | Type of pyrolysis | Reactor | Pyrolysis conditions | Products yield (%) | Reference |
|---|---|---|---|---|---|
| Mulberry branches | Slow | Fixed bed – electrically heated furnace | 500 °C; SRT: 120 min; HR: 2° C/min | Biochar: 33<br>PA: 37<br>Gas: 30 | Cheng et al. (2022) |
| *Rosmarinus officinalis* leaves | Slow | Fixed bed – electrically heated furnace | 310 °C; 200 W; SRT: 170 min | Biochar: 10<br>PA: 71<br>Bio-oil: 10<br>Gas: 9 | Ma et al. (2013) |
| | | | 440 °C; 700 W; SRT: 90 min | Biochar: 8<br>PA: 59<br>Bio-oil: 14<br>Gas: 19 | |
| | | | 530 °C; 1200 W; SRT: 80 min | Biochar: 6<br>PA: 67<br>Bio-oil: 9<br>Gas: 18 | |
| *Eucommia ulmoides* Oliver branches | Slow | Fixed bed (stainless steel) - electrically heated furnace | 650 °C; SRT: 120 min; HR: 2 °C/min | Biochar: 45.6<br>PA: 27.0<br>Gas: 27.4 | Xue et al. (2022) |
| Cottonseed | Intermediate | Fixed bed – electrically heated furnace | 500 °C; SRT: 10 min; HR: 30 °C/min | Biochar: 35<br>Bio-oil: 51<br>Gas: 14 | Rajamohan et al. (2022) |
| Date palm leaves (*Phoenix dactylifera*) and mesquite plants (*Prosopis juliflora*) | Slow | Fixed bed - coupled to a muffle furnace | 600 °C; SRT: 120 min | Biochar: 26.7 | Al-Rabaiai et al. (2022) |
| Apricot stone (*Prunus armeniaca*) | Intermediate | Fixed bed (Jenkner type retort) – electrically heated furnace | 1096 °C; SRT: 30 min; HR: 278 °C/min | Biochar: 34.7<br>Bio-oil: 35.8<br>Gas: 29.5 | Özçimen and Ersoy-Meriçboyu (2010) |
| Hazelnut shell (*Corylus avellana*) | | | 1023 °C; SRT: 30 min; HR: 278 °C/min | Biochar: 51.5<br>Bio-oil: 42.0<br>Gas: 6.5 | |
| Grapeseed (*Vitis vinifera*) | | | 1023 °C; SRT: 30 min; HR: 278 °C/min | Biochar: 53.0<br>Bio-oil: 39.1<br>Gas: 7.9 | |
| Chestnut shell (*Castanea sativa* Mill) | | | 1023 °C; SRT: 30 min; HR: 278 °C/min | Biochar: 58.8<br>Bio-oil: 36.1<br>Gas: 5.1 | |
| Exhausted grape marc | Slow | Fixed bed (Vertical tubular reator with internal drawers) – electrically heated furnace | 500 °C; SRT: 60 min; HR: 5 °C/min | Biochar: 33.8 | Ferjani et al. (2019) |

**Table 2** (continued)

| Biomass | Type of pyrolysis | Reactor | Pyrolysis conditions | Products yield (%) | Reference |
|---|---|---|---|---|---|
| Waste tobacco stem | Slow | Horizontal tube furnace | 600 °C; SRT:60 min; HR: 10 °C/min | Biochar: 31,8<br>Bio-oil: 37,1<br>Gas: 31,1 | Ma et al. (2023) |
| Date seeds leftover | Intermediate | Tubular reactor (quartz) - horizontal furnace | 500 °C; SRT:30 min; HR: 10 °C/min | Biochar: 2,7<br>Bio-oil: 22,1<br>Gás: 75,1 | Al-Mardeai et al. (2023) |
| Giant reed stems (*Arundo donax* L.) | Slow | Vacuum tube furnace | 600 °C; SRT: 120 min | Biochar: 30.5 | Wang et al. (2013) |
| Rice husk | Slow | "Engineered equipment" (patent) | 500 °C; SRT: 60 min | Biochar: 35 | Miao et al. (2023) |
| Winery composted waste | Slow | Fixed bed (Quartz) – electrically heated furnace | 700 °C; SRT:120 min; HR: 10 °C/min | Biochar: 28 | Silva et al. (2021) |
| Rice straw | Intermediate | Fixed bed – electrically heated furnace | 600 °C; SRT: 20 min; HR: 20 °C/min | Biochar: 34.9 | Gong et al. (2019) |
| *Populus grimminge* (Grimminge poplar) | Intermediate | Fixed bed – electrically heated furnace | 723 °C; HR: 30 °C/min | Biochar:361<br>PA: 42<br>Gas: 27 | Stals et al. (2010) |
| *Salix jorunn* ("Jorunn" willow) | | | | Biochar:22<br>PA: 45<br>Gas: 33 | |
| *Salix fragilis* (crack willow) | | | | Biochar:23<br>PA: 52<br>Gas: 25 | |
| Palm kernel shell | Fast | Entrained-flow reactor - electrically heated furnace | 600 °C; VRT: 5–6 s | Biochar: 20,89<br>PA: 73,74<br>Gas:5,37 | Maliutina et al. (2017) |
| | | | 900 °C; VRT: 5–6 s | Biochar: 14,89<br>PA: 53,33<br>Gas: 31,78 | |
| Pine sawdust | Intermediate | Tubular with moving biomass crucible | 550 °C; SRT: 15 min | Biochar: 27<br>PA: 43<br>Gas: 30 | Cen et al. (2023) |
| *Prosopis juliflora* | Slow | Fixed bed (retort) – electrically heated furnace | 400 ºC | Biochar: 23<br>PA: 33<br>Gas: 43 | Prabhahar et al. (2022) |
| Wood wastes | Fast | Fluidized bed | 500–650 °C; VRT: <2 s; HR: 105 ºC/s | | Zhang et al. (2018) |

SRT: Solids residence time; VRT: Vapor residence time HR: Heating rate; PA: Pyroligneous acid

Guedes et al., 2018; Xu et al., 2011). Vapor residence times are typically disclosed in studies involving fast or intermediate pyrolysis. In fluidized bed systems, vapor residence times can be easily manipulated by adjusting the carrier gas flowrate and biomass feed rate, and thus, this parameter is often estimated or discussed. On the other hand, solids residence time and heating rate are challenging to measure or estimate in fluidized bed systems and are frequently undisclosed.

Various designs of pyrolysis reactors are available, including fixed bed, fluidized bed, circulating fluidized bed, ablative, microwave, rotating cone, auger, vacuum and solar reactor (Guedes et al., 2018; Lewandowski et al., 2019; Rahimi et al., 2022). The operation, advantages, and disadvantages of each reactor type are presented in Table 3.

The different types of reactors mainly influence product yield, heat exchange, process time, and cost. Pyrolysis system designs range from simple to complex structures, typically including a feeder, reactor, condenser, and liquid collector. Equipment can accommodate a range of particle sizes, although some may have limitations for large particles. Various heat transfer options (usually indirect) are available, such as electric furnaces, oil, gas, or biomass furnaces, which are placed around the reactor, transferring heat through the reactor walls. Heat carriers, such as sand, glass, or steel particles, can transport heat from the heat source to the pyrolyzed material. These characteristics define the range of operational parameters and directly impact the construction and operation costs.

Fluidized bed and fixed bed reactors are the most utilized and researched for biomass fast pyrolysis due to their simple construction and ease of operation (Dhyani & Bhaskar, 2018). Fixed bed reactors are typically operated in batch mode. They find applications from laboratory to industrial scales due to their structural simplicity, which is linked to the absence of mechanized feeding and removal systems. Continuous pyrolysis systems inherently involve complexity, requiring biomass feeding and biochar removal systems that enable solid flow without permitting vapor leakage. In a fluidized bed, a carrier gas is used to fluidize a bed of solid particles, ensuring excellent heat transfer and also dragging pyrolyzed particles out of the reactor, eliminating the need for mechanical transport or mixing. Additionally, fluidized-bed reactors are favorable for fast pyrolysis due to their ability to rapidly pyrolyze biomass particles (high heat transfer rates), and control residence time through the manipulation of feed-rate and carrier gas flowrate. Additionally, fluidized-bed reactors are advantageous for fast pyrolysis due to their capacity to rapidly pyrolyze biomass particles (high heat transfer rates) and control residence time through the manipulation of feed-rate and carrier gas flowrate. Pyrolysis systems can utilize vacuum to reduce vapor residence times and prevent secondary cracking of pyrolysis products, increasing the system's complexity.

In this regard, factors such as the type of reactor, experimental process conditions, equipment configuration, and the biomass directly influence product yields, their physical-chemical properties and applications. Throughout this section, it becomes evident that extensive research has been dedicated to exploring the variables associated with pyrolysis, aiming to obtain desired products. Therefore, the pyrolysis of different agricultural wastes emerges as a rich research topic, offering numerous challenges and opportunities. This perspective not only highlights its potential as a tool to enhance efficiency in waste conversion but also underscores its significant contribution to sustainable practices and the advancement of the circular economy in the waste management context.

**Table 3** Types of pyrolysis reactors and operation, and their respective advantages and disadvantages

| Reactor | Operation/Configuration | Advantages | Disadvantages | References |
|---|---|---|---|---|
| Fluidized bed | Continuous operation. The bed is fluidized by gas flow, which maximizes mixing and interaction between gas and particles. Heating is done by external combustion, by direct transfer (adding hot solids to the bed), or indirect (hot gas or steam passing through tubes in the bed). Classification: bubbling, entrained, and circulating. | → Simple design;<br>→ Easy scale-up;<br>→ Ability to control residence time;<br>→ High heat and mass transfer coefficients;<br>→ Uniform heat transfer. | → High cost of construction and operation;<br>→ External heating;<br>→ Continuous makeup of bed material required;<br>→ Reactor degradation by particle collision. | Dhyani and Bhaskar (2018); Lewandowski et al. (2019); Cai et al. (2021); Al-Rumaihi et al. (2022) |
| Fixed bed | The static reaction bed can be composed only of the raw material or mixed with sand or in the presence of a catalyst. The process may occur in the presence or absence of inert gas. The reactor can be vertical or horizontal. The central part of the reactor is a heating chamber, where the main heat flow is provided through a fuel burner. | → Simple construction;<br>→ Easy operation;<br>→ Low cost;<br>→ Applicability of raw material;<br>→ High yield of biochar. | → Heat and mass transfer problem;<br>→ Insufficient heat transfer;<br>→ Scalar inadequacy;<br>→ Continuous feeding is difficult;<br>→ Small capacity;<br>→ Low liquid and gas yield. | Lewandowski et al. (2019); Cai et al. (2021); Al-Rumaihi et al. (2022) |
| Ablative | Thermal degradation of biomass occurs by pressure against the hot wall of the reactor. The reaction rate is highly influenced by different factors such as the pressure, the relative velocity of the biomass on the heating surface and the surface temperature of the reactor. | → It has no requirement on the size of the particles, thus allowing for a large raw material;<br>→ Saves costs and energy with pre-treatment of biomass;<br>→ Allows good mechanical abrasion of coal. | → Complex structure, due to erosion against the reactor wall;<br>→ The morphology, size and flow characteristics of the particles represent a significant constraint. | Dhyani and Bhaskar (2018); Cai et al. (2021); Al-Rumaihi et al. (2022) |
| Rotary cone | Biomass and heat carriers move in a spiral along the reactor wall due to centrifugal force in the continuous rotation process. Heat transfer occurs by direct contact during the mixing of biomass and heat carriers. | Carrier gas is not required, which reduces operating costs.<br>→ Instant heating of biomass due to high heat transfer;<br>→ Efficiency is relatively high;<br>→ Shorter residence time. | → Complex structure, higher construction cost. | Dhyani and Bhaskar (2018); Cai et al. (2021) |

**Table 3** (continued)

| Reactor | Operation/Configuration | Advantages | Disadvantages | References |
|---|---|---|---|---|
| Auger/ screw | Continuous tubular reactors, in which solid biomass is transported by a rotating screw. In addition to mixing the feed, the screw controls the residence time of the biomass in the reactor. Suitable for conveying granules and powder materials. Heat transfer is along the tubular wall of the reactor. | → It can be built compactly or portable, allowing use at the biomass generation site or where biomass is disposable in abundance. Thus, it reduces operating and transportation costs;<br>→ Demands less energy in the pre-treatment of the raw material;<br>→ Anti-winding and anti-clogging;<br>→ A large amount of biomass. | → Lower heat transfer rates;<br>→ Relatively longer residence times;<br>→ Mechanical wear. | Dhyani and Bhaskar (2018); Cai et al. (2021); Campuzano et al. (2019) |
| Microwave | Thermal decomposition is facilitated by the thermal energy applied by the microwave. The energy generated is effect of the atomic or molecular interaction of the material and the electromagnetic field. Heat transfers occur from the internal surface to the external surface of the material. | → Prevents the formation of secondary reaction;<br>→ Better product quality;<br>→ Short reaction time;<br>→ High heating efficiency, fast heating, uniform and bulky. | → Dependence on the dielectric properties of the biomass;<br>→ Difficulty in mixing and controlling particle size;<br>→ Difficult to scale up;<br>→ Requires microwave compatible materials. | Dhyani and Bhaskar (2018); Lewandowski et al. (2019); Al-Rumaihi et al. (2022); Morgan et al. (2017) |
| Vacuum | The biomass is transported through a belt and mechanically agitated. | → Carrier gas is not required. | → Limitation for larger particles;<br>→ Complicated structure and maintenance;<br>→ Risk of air intake;<br>→ Requires special equipment for feeding. | Al-Rumaihi et al. (2022) |
| Solar | Energy is provided by solar radiation concentrated on the outer walls of the reactor. | → Lower cost of operation;<br>→ Low heating costs;<br>→ Source of renewable energy. | → Need to investigate an adequate solar project. | Lewandowski et al. (2019); Al-Rumaihi et al. (2022) |

## 4 Obtaining products from the pyrolysis of agricultural wastes

The pyrolysis of agricultural wastes has garnered significant attention from researchers worldwide. The applications of products obtained through pyrolysis are diverse, with a particular focus on bioenergy for bio-oil and gases, and remediation for biochar. On the other hand, alternative uses in the agricultural, food, and pharmaceutical sectors are growing and promising, as shown in the following sections.

### 4.1 Solid fraction from pyrolysis: characteristics and applications

Biochar, the solid product from pyrolysis, is primarily composed of carbon (63–91%) and oxygen (10–30%) (Guedes et al., 2018; Yadav et al., 2023). During biochar production, there is an increase in the surface porosity of the material, accompanied by the formation of various functional groups, which play a crucial role in the presence of surface charges. The negatively charged sites on the biochar's surface represent areas where cations can bind and be exchanged, resulting in a high cation exchange capacity (CEC) (Ghodake et al., 2021). In pyrolysis processes, the yields of the solid fraction decrease with increasing heating rate and pyrolysis temperature. This is attributed to the greater primary degradation of biomass at higher temperatures or through the secondary decomposition of char residue (Demiral & Ayan, 2011; Ghodake et al., 2021; Li et al., 2021; Yadav et al., 2023). The elevation of the pyrolysis temperature tends to remove oxygen and hydrogen from the solids through dehydration and deoxygenation reactions, reducing the O/C and H/C ratios. This process makes the biochar more hydrophobic, decreases the cation exchange capacity (CEC), and increases the surface porosity (Ghodake et al., 2021; Li et al., 2021; Uchimiya et al., 2013; Yadav et al., 2023).

The variability in the physicochemical properties of biochar (surface area, porosity, pH, polarity, functionalities, and elemental composition) provides a means to optimize its effectiveness in specific applications. The quality and performance of this material significantly vary depending on the types of feedstocks and pyrolysis conditions, primarily temperature (Dhyani & Bhaskar, 2018; Ghodake et al., 2021; Oliveira et al., 2017; Venkatachalam et al., 2023; Yadav et al., 2023). Additionally, pre- and post-treatments (physical and chemical techniques) can be applied to enhance the properties and characteristics of biochar. Physical modifications primarily occur through grinding processes that reduce the size of biochar and the introduction of steam that increases the surface area and porosity through surface oxidation of biochar (Cho et al., 2023; Fahmi et al., 2018; Venkatachalam et al., 2023). On the other hand, chemical modifications influence both physical properties (area and porosity) and chemical (desired functional groups) and electronic (structure modulation) properties (Cho et al., 2023; Venkatachalam et al., 2023). Chemical treatments include activation, doping with inorganic compounds, magnetization, among others (Sun et al., 2023; Xu et al., 2019; Zhu et al., 2018). The properties of biochar produced from the pyrolysis of agricultural residues and its modifications play a crucial role in its various applications, as outlined in Table 4.

Biochar finds extensive use in environmental remediation (see Table 4) due to its strong interaction with various contaminants in aqueous, gaseous, or solid phases (Gwenzi et al., 2021; Oliveira et al., 2017; Yadav et al., 2023). The removal of contaminants using biochar occurs through the interaction of its functional groups (COOH, OH, R-OH, $NH_2$) with the compounds, involving mechanisms such as precipitation, complexation, ion exchange, and adsorption (Cho et al., 2023; Gwenzi et al., 2021; Oliveira et al., 2017). The predominant mechanisms include chemisorption (electrophilic interaction) and physisorption (hydrogen and electrostatic bonds, π-π interactions) (Cho et al., 2023; Deng et al., 2017; Gwenzi et al., 2021; Oliveira et al., 2017; Zhang et al., 2017; Zhu et al., 2018). Biochar's physical properties, such as surface area and porosity, enhance the removal of contaminants by providing a higher number of active sites that increase the interaction between biochar and contaminants.

**Table 4** Applications of biochar produced from different agricultural wastes

| Biomass | Application | Main results | Referência |
|---|---|---|---|
| Rice husk | Nitrogen conversion and retention during composting | Accelerated the decomposition of organic matter while reducing nitrogen and carbon loss | Wang et al. (2023) |
| Woody biomass *of* date palms (*Phoenix dactylifera*) and mesquite trees (*Prosopis juliflora*) | Effect of biochar on microbial count and soil respiration | Less impact on soil microbes, high organic content, surface area, cation exchange capacity and thermostability | Al-Rabaiai et al. (2022) |
| Apricot kernel (*Prunus Armenian*) | Industrial applications as an alternative to solid fossil fuels and a carbon source* | Fixed carbon content (71.7%) and calorific value (30.76 MJ/kg) | Özçimen and Ersoy-Meriçboyu (2010) |
| Hazelnut husk (*Corylus hazelnut*) | | Fixed carbon content (63.16%) and calorific value (29.08 MJ/kg) | |
| Grape seed (*Vitis vinifera*) | | Fixed carbon content (50.96%) and calorific value (26.73 MJ/kg) | |
| Chestnut husk (*Castanea sativa* Mill) | | Fixed carbon content (60.32%) and calorific value (25.86 MJ/kg) | |
| Waste of exhausted grape marc | Soil corrective and/or as a pollutant adsorbent* | Interesting amount of minerals (P, Ca, Mg, Si, Fe, Al, Na and K), as well as a surface area (253 $m^2/g$) and microporosity (129 µm) | Ferjani et al. (2019) |
| *Canna indica* | Adsorption of Cd(II) from aqueous solution | $Q_m$ was 188.8 mg/g (Langmuir model). Mechanisms: precipitation, ion exchange of metals, complexation with functional groups containing surface oxygen and coordination with π electrons | Cui et al. (2016) |
| Hardwood | Removal of Cu(II) and Zn(II) from aqueous solution | $Q_m$ for Cu(II) and Zn(II) were 6.79 and 4.54 mg/g (Langmuir model), respectively | Chen et al. (2011) |
| Corn straw | | $Q_m$ for Cu(II) and Zn(II) were 12.52 and 11 mg/g (Langmuir model), respectively | |
| Seagrass (*Spartina alterniflora*) | Removal of Cu(II) from aqueous solution | $Q_m$ was 48.49 mg/g (Langmuir model). Mechanism: ion exchange, π–metal interaction, precipitation | Li et al. (2013) |
| Giant reed stalks (*Arundo donax L.*) | Reduce $N_2O$ emissions from agricultural soil | The emission reduction was attributed to the increase in N immobilization and the decrease in denitrification in soils amended with biochar | Wang et al. (2013) |
| Bamboo waste | Evaluate the influence of biochar on the leaching and bioavailability of pentachlorophenol (PCP) in soil | Biochar decreased cumulative PCP leaching by 42.3% and also reduced PCP concentrations by 56 and 65% in methanol and distilled water extracts, respectively | Xu et al. (2012) |
| Rice husk | Soil conditioner | Biochar improved both rice biomass production and resistance to lodging by promoting the co-deposition of silicon with hemicellulose and lignin in cell walls. It also regulated key genes involved in cell wall synthesis and rice growth | Miao et al. (2023) |

**Table 4** (continued)

| Biomass | Application | Main results | Referência |
|---|---|---|---|
| Miscanthus straw pellets | Soil conditioner to reduce the absorption of heavy metals mobilized with organic acid in soils | Reductions: Cd: 17–31% (malic acid) and 22–65% (oxalic acid); Cu: 11–36% (citric acid) and 21–39% (malic acid) and 0–19% (oxalic acid); Pb: 16% (citric acid) | Achor et al. (2020) |
| Rice straw | Evaluate the effect of dissolved organic matter derived from biochar on the adsorption of sulfamethoxazole (SMX) and chloramphenicol (CAP) | Dissolved organic matter shortened the equilibrium time of SMX and CAP adsorption; Higher biochar adsorption capacity with more organic matter dissolved for SMX, and decreased for CAP | Yang et al. (2020) |
| Mixture of rapeseed straw and eggshell | Phosphorus adsorption | $Q_m$ was 103.9 mg/g (Langmuir model); Mechanisms: Ca-P precipitation, hydrogen bonding, electrostatic attraction | Cao et al. (2020) |
| Pure alkali spent coffee grounds | Adsorption of micropollutants (carbamazepine, 17 α- ethinyl estradiol, ibuprofen) with raw and alkali-modified biochar | The highest removals were by alkali-modified biochars (44.5–96.1%) compared to untouched biochars (2.5–18.5%); Alkaline modification can enhance surface structural properties of biochars; Mechanism: chemosorption (electrostatic and hydrophobic interactions); | Shin et al. (2020) |
| Winery composting waste | Adsorption of ketoprofen and 2-nitrophenol | $Q_m$ were 376 and 119 mg/g for 2-nitrophenol and ketoprofen, respectively (Sips model) | Silva et al. (2021) |
| Corn cob | Adsorption of 2,4-dichlorophenoxyacetic (2,4-D) | $Q_m$ was 37.40 mg/g (Langmuir model). Mechanism: pore filling and chemical adsorption (π-π interactions (electron-donor-acceptor) and hydrogen bonding). | Binh and Nguyen (2020) |
| Rice straw | Adsorption-desorption of nutrients ($NH_4^+$ -N, $NO_{3_}$ -N, P, K) | The adsorption capacities of the nutrients were between 13–17 mg/g. After five adsorption-desorption cycles, nutrient adsorption was still > 40% of the first adsorption. Biochar has a relatively high reuse. | Gong et al. (2019) |

*: Potential application; $Q_m$ Maximum adsorption capacity

Biochar exhibits high potential for removing both organic and inorganic contaminants. In terms of organic contaminants, studies have shown that biochar is effective in removing agrochemicals (Deng et al., 2017; Herath et al., 2016; Silva et al., 2021), pharmaceuticals (Silva et al., 2021; Sun et al., 2023), polycyclic aromatic hydrocarbons (PAHs) (Zhu et al., 2018), and dyes (Guan et al., 2023; Li et al., 2016; Xu et al., 2019). Biochar is also associated with the removal of inorganic contaminants such as heavy metals (Fahmi et al., 2018; Sun et al., 2023), ammonia, nitrate, and phosphate (Wang et al., 2020; Yao et al., 2012). Gas treatments reported in the literature using biochar include the removal of mercury (Zhu et al., 2016), ozone (Zhou et al., 2018), carbon dioxide (Xu et al., 2019), nitrogen oxides (Zhu et al., 2016), sulfur dioxide and hydrogen sulfide (Braghiroli et al., 2019; Das et al., 2019), as well as volatile organic compounds (VOCs) (Zhang et al., 2017). This broad spectrum of treatment capacity underscores the versatility and efficiency of biochar in addressing various types of contaminants.

The incorporation of biochar into soil has gained prominence due to its advantageous characteristics, which enhance various soil properties. Biochar exhibits significant stability

and resistance to degradation compared to its raw materials, reducing the bioavailability of contaminants in the soil. The addition of biochar to coarse or sandy-textured soils has demonstrated a reduction in $N_2O$ emissions and $NO_3$- leaching (see Table 4), thereby mitigating soil nitrogen losses (Borchard et al., 2019; Matos et al., 2021; Patel & Panwar, 2023; Woolf et al., 2010). According to Woolf et al. (2010), applying biochar to the soil could potentially decrease annual $CO_2$, $N_2O$, and $CH_4$ emissions by 12% without compromising food security and habitat conservation. The capacity to prevent carbon conversion into atmospheric $CO_2$, known as carbon sequestration, contributes to climate change mitigation by reducing greenhouse gas (GHG) emissions (Borchard et al., 2019; Cabrera et al., 2014; Matos et al., 2021). Furthermore, carbon sequestration by biochar maintains carbon in its most stable form in the soil, playing a crucial role in the healthy functioning of the terrestrial ecosystem by influencing various soil processes and properties such as fertility, water retention, and structure.

Biochar significantly influences soil responses to water, aggregation, aeration, elasticity, and permeability by enhancing fundamental soil physical properties like structure, density, and texture. Consequently, it has the ability to retain rain and irrigation water for gradual release during dry periods, and can also regulate soil pH by retaining and releasing $H^+$ and $OH^-$ ions. Notably, biochar facilitates the soil structure through chemical bonds between the material and inorganic macromolecular structures. This helps reduce soil salinity by retaining metal ions, serving both as essential plant nutrients (e.g., calcium, magnesium, iron, copper) and as potential toxins (e.g., aluminum) (Borchard et al., 2019; Cabrera et al., 2014; Matos et al., 2021). Consequently, biochar emerges as a valuable soil amendment, enhancing plant growth yield, improving soil fertility, and fostering sustainable agricultural productivity by promoting the activity of beneficial microorganisms (Ronix et al., 2021; Yang et al., 2021; Patel & Panwar, 2023).

The need for increasing food production has led to the growing utilization of synthetic fertilizers, which pose potential pollution risks (Patel & Panwar, 2023). Therefore, actions are needed to reduce the environmental impact and improve economic aspects. Biochar is studied as a potential solution to control the release of fertilizers and enhance their potential. It serves as a support for these compounds, reducing their loss, costs and GHG emissions. The internal surfaces of biochar are soaked with an organic coating, increasing water storage capacity and nutrient exchange capability. Thus, it can result in more efficient plant growth of different crops, improve soil correction and nutrient recycling from agro-industrial waste (Schmidt et al., 2019; Xu et al., 2011). Another use in the agro-industrial context is the incorporation of biochar in ruminant feed to avoid the absorption of toxic substances and prevent diseases, as this material can adsorb various types of contaminants, poisons, and gases from the ruminant digestive tract (Pereira et al., 2014).

Biochar can be applied to composting processes, bringing a series of positive effects, such as: (a) improvement of the microbiological action and acceleration of the process; (b) significant retention of nutrients and humidification; (c) reduction of GHG emissions; (d) immobilization of heavy metals (Guo et al., 2020; Nguyen et al., 2022; Sánchez-Monedero et al., 2018; Wang et al., 2023). Wang et al. (2023) simulated the implementation of biochar in composting systems. The addition of biochar significantly improved the efficiency of composting, increased the $NO_3^-$-N concentration, and reduced the nitrogen loss rate. Reducing nitrogen loss decreases the release of $NH_3$, $N_2O$ and $NO_X$ which cause negative environmental impacts. Moreover, composting modifies the structure and properties of the pores of the biochar, causing an increase in the size of micropores and, consequently, a larger

surface area. Regarding surface chemistry, this process increases the CEC and formation of inorganic phases in the pores of biochar that affect the interaction with compounds present in the composting (Nguyen et al., 2022; Sanchez-Monedero et al. 2018). Therefore, the incorporation of biochar into the composting represents a beneficial synergistic interaction between agriculture, rural ecology, and economy.

Biochar can serve as a solid biofuel in industrial applications, offering an alternative to solid fossil fuels (Dhyani & Bhaskar, 2018; Kakku et al., 2023; Ma et al., 2023; Özçimen & Ersoy-Meriçboyu, 2010; Rahimi et al., 2022). It can be burned to provide heat for the pyrolysis process, increasing thermal efficiency and reducing costs (Dhyani & Bhaskar, 2018). Özçimen and Ersoy-Meriçboyu (2010) produced biochar from various residues with high calorific value (25–31 MJ/kg) and high fixed carbon content (50–72%). In a study of tobacco stem residues, Ma et al. (2023) obtained biochar with a carbon content ranging from 57 to 60% and a calorific value of 17 to 19 MJ/kg. Kakku et al. (2023) synthesized biochars in a continuous auger reactor, resulting in materials with calorific values of 24.15 MJ/kg for cotton stalks and 24.57 MJ/kg for mustard husk. These studies urderscore the potential use of biochar from agro-waste as an alternative solid biofuel. Considering its diverse applications, biochar has the potential to positively impact several Sustainable Development Goals (SDGs) (Rempel et al., 2021). Therefore, biochar, with its controllable properties through pyrolysis conditions and enhancements through pre- and post-treatments, is a promising material for applications in environmental, agricultural, and energy areas (Fig. 2).

### 4.2 Gaseous fraction from pyrolysis: characteristics and applications

Non-condensable gases produced during biomass pyrolysis consist primarily of hydrogen ($H_2$), carbon dioxide ($CO_2$), carbon monoxide (CO), ethane ($C_2H_6$), propane ($C_3H_8$), and methane ($CH_4$) (Demiral & Ayan, 2011; Foong et al., 2021; Ma et al., 2023). The specific composition of these gases depends on the biomass being utilized. Hemicellulose is responsible for a higher $CO_2$ yield due to its carboxyl content. Conversely, cellulose generates a higher yield of CO, primarily due to the thermal cracking of carbonyl and carboxyl groups. The pyrolysis of lignin results in higher releases of $H_2$ and $CH_4$, attributed to the greater

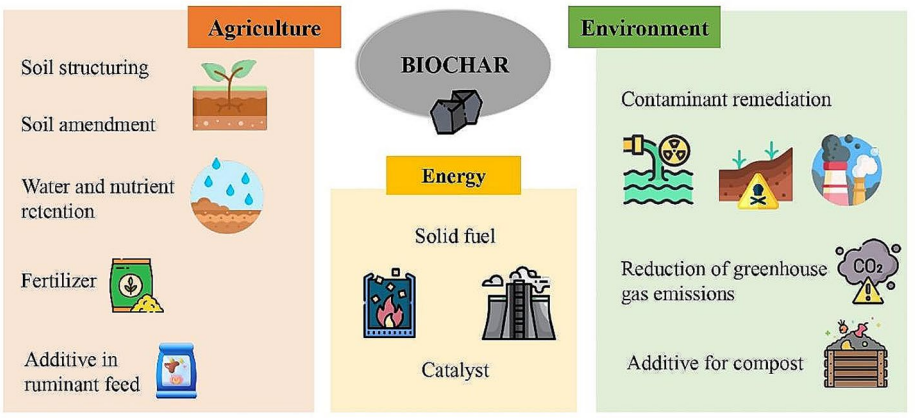



**Fig. 2** Applications of biochar produced from the pyrolysis of agricultural wastes

presence of aromatic rings and methoxyl functional groups (Dhyani & Bhaskar, 2018; Hubble & Goldfarb, 2021).

The investigation of the recovery of these gaseous contents is associated with the evaluation of different conditions (mainly temperature) and the evolution of gases, where higher yields of gaseous products are desirable (Foong et al., 2021; Hubble & Goldfarb, 2021; Prabhahar et al., 2019). Higher temperatures cause greater production of non-condensable gases, attributed to the increase in the rate of secondary cracking reactions of volatiles. Further decomposition of biochar at higher temperatures can also generate non-condensable gas products (Demiral & Ayan, 2011; Hubble & Goldfarb, 2021; Prabhahar et al., 2019; Xu et al., 2009). Kakku et al. (2023) achieved a gas yield of 55% from the pyrolysis of mustard husk in a continuous auger reactor at 700 °Ma et al. (2023) studied the nitrogen conversion from the slow pyrolysis of tobacco stem at different temperatures (400–800 °C, obtaining higher gas yields (36.6–40%) at higher temperatures (700–800 °C). In addition to the increased yield, the temperature also led to an increase in the calorific value of the gases, ranging from 39.7 MJ/(Nm³) at 400 °C to 68 MJ/(Nm³) at 800 °C. The authors also observed that $H_2$ yields increased from 4.66% at 400 °C to 37.53% at 800 °C. These studies highlight that gas production is favored at high temperatures.

Pyrolysis gases are considered a fuel gas used for energy and heat generation. They can be applied in various types of equipment for industrial combustion purposes, serving as a heat source for the pyrolysis process itself, or utilized in internal combustion engines and gas turbines to produce electricity (Dhyani & Bhaskar, 2018; Guedes et al., 2018; Ma et al., 2023; Rahimi et al., 2022; Xu et al., 2009). Xu et al. (2009) studied the effect of pyrolysis temperature on gaseous products, aiming at their energetic use in the process itself. If the temperature is too high (>650 °C), there is excess gas produced, and if it is too low (<500 °C) not enough gas is produced to sustain the process. The ideal temperature range for the gas to fulfill all the process heat demand was 550–600 °C (Xu et al., 2009). Pyrolysis gas as a fuel can be less harmful to the environment than fossil fuels, resulting in lower emissions of CO and ozone into the atmosphere due to the presence of $H_2$, CO and $CO_2$, and the absence of sulfur (Imam & Capareda, 2012; Ma et al., 2023; Prabhahar et al., 2019).

The recovery of enriched gases, such as hydrogen gas ($H_2$), has been studied for various applications. It can be utilized in ammonia and methanol production and serves as a carbon-free alternative fuel (Foong et al., 2021; Hubble & Goldfarb, 2021; Prabhahar et al., 2019; Rahimi et al., 2022). Prabhahar et al. (2019) investigated the production of $H_2$ through thermal and catalytic pyrolysis of rice husks using nickel as a catalyst. Catalytic pyrolysis achieved higher $H_2$ production (33.33%) compared to thermal pyrolysis (21.67%) at 400 °C. According to Dhyani and Bhaskar (2018), the gaseous products depend on the chemical structures of the raw materials. Thus, depending on the raw material used for carbonization, toxic compounds may be generated, requiring control to prevent their release into the environment. Raw materials containing chlorine and sulfur in their composition can favor the formation and release of organochlorines, such as poly-chlorinated-dibenzo-furans and poly-chlorinated-dibenzo-p-dioxins (PCDF and CPDP, respectively) (Al-Rumaihi et al., 2022; Uzoejinwa et al., 2018). These gases can harm human health and lead to complications such blindness and vomiting (Al-Rumaihi et al., 2022; Uzoejinwa et al., 2018). The thermal decomposition of chlorinated plastic waste, such as PVC, which is sometimes present in mixed farm wastes, can also result in the formation of such chlorinated compounds (Al-Rumaihi et al., 2022; Uzoejinwa et al., 2018).

Biomasses, which generally do not contain appreciable amounts of chlorine, are considered safe for pyrolysis and gas production. This is in contrast to other types of waste, such as municipal waste, which may contain chlorinated plastics leading to the production of toxic gases (Foong et al., 2021). It is crucial to characterize the generated gases to obtain detailed information about their composition and prevent the potential release of toxic substances, thereby demonstrating a commitment to environmental safety.

### 4.3 Liquid fraction from pyrolysis: characteristics and applications

Pyroligneous extract (PE) can be defined as a complex mixture of compounds captured as liquids after the thermal degradation of organic matter. These compounds primarily result from the degradation of cellulose, hemicellulose, lignin, and others components, including phenolic compounds (Hu et al., 2023; Wu et al., 2015). PE has a composition rich in carbon and a significant concentration of high-value-added chemicals, obtainable through the fractionation of the liquid. Since pyrolysis involves minimal deoxygenation, PE maintains the same elemental composition as the initial biomass (Alhazmi & Loy, 2021; Singh et al., 2023). However, the overall amount of each component in lignocellulosic biomass is critical in determining how effectively the biomass can be transformed into green fuels or valuable chemicals (Singh et al., 2023). The composition, yield and ratio between liquid phases of pyrolysis vary depending on process parameters, such as the chemical composition of the feedstock, moisture content, particle size, temperature, heating rate, and residence time (Cai et al., 2024; Gollakota et al., 2023; Güllü & Demirbaş, 2001; Morvová et al., 2003; Ma et al., 2023).

The liquid fraction resulting from pyrolysis can be separated into two distinct phases: an aqueous phase (also known as pyroligneous acid) and a denser oily organic phase, commonly referred to as bio-oil. It is important to note that some authors may refer to both phases as "bio-oil", and in certain cases, liquid extracts may not readily undergo phase separation, such as in flash pyrolysis processes (Bridgwater, 2012). Typically, the primary constituent of pyrolysis extracts is water, accompanied by more than 200 organic compounds (Gollakota et al., 2023; Loo et al., 2008; Mungkunkamchao et al., 2013; Wu et al., 2015). These compounds result from the thermal decomposition of lignocellulosic biomasses, which are structurally composed of three main polymers: cellulose (30–60%), hemicellulose (20–40%) and lignin (15–30%) (Yin et al., 2021). Cellulose and hemicellulose share similar structures and characteristics, with initial degradation occurring in the range of 200 to 400 °C, leading to the production of alcohols, aldehydes, ketones, acids, furans, sugars. On the other hand, lignin is primarily decomposed mainly into phenols and some aromatics in the range of 250 to 500 °C (Hu et al., 2023; Chai et al., 2024).

Pyroligneous extract can be utilized for heat and electricity generation (Silos-Llamas et al., 2023). With a low percentage of $CO_2$ produced, it is considered a cleaner fuel compared to fossil fuels, emitting approximately 50% less nitrogen oxide than diesel oil (Singh et al., 2023). Despite numerous advantages over heavy fuel oil, such as higher lubricity, lower toxicity, and greater biodegradability, pyroligneous extract is an acidic (pH 3–4), viscous or free-flowing, dense, polar, dark-colored liquid with a high oxygen content (30–45%) and low stability, resulting in a relatively low calorific value. Its high water content (>26% by weight) is viewed as a disadvantage for use as fuel (Alhazmi & Loy, 2021; Gollakota et al., 2023; Gholizadeh et al., 2019; Kassim et al., 2022; Singh et al., 2023). Therefore,

its direct use is limited, necessitating improvements in its physical-chemical properties for applications in engines, boilers, and chemical products (Alhazmi & Loy, 2021; Cai et al., 2024; Gollakota et al., 2023; Singh et al., 2023). Pyroligneous extract has also demonstrated potential in the production of high-value chemical products, suitable for extraction, recovery, or synthesis of preservatives, food flavors, adhesives, agrochemicals, resins, and fertilizers (Al-Mardeai et al., 2023; Cai et al., 2024; Singh et al., 2023).

The aqueous phase of PE consists of numerous low-molecular-weight organo-oxygenated compounds, while the non-aqueous phase (organic) comprises organics with low solubility in water, especially high-molecular-weight aromatics (Güllü & Demirbaş, 2001). As noted by Güllü and Demirbaş (2001), the aqueous phase has higher concentration of acetic acid, methanol and acetone, compared to the organic phases. In fractional pyrolysis, the liquid fraction formed at temperatures ranging from room temperature to 120–150 °C exhibits a yellowish to brown-red color, with no detectable oil phase. At higher temperatures, the liquids become darker, more organically characteristic, and may contain a denser bio-oil phase rich in oxygen (Cai et al., 2024; Guedes et al., 2018; Gollakota et al., 2023). After the extraction process, polymerization reactions occur among the liquid compounds until stabilization, a maturation process that can extend up to two years (Mohan et al., 2006). Once stabilized, the organic and aqueous phases can be separated by decantation or distillation, depending on the intended application (Mohan et al., 2006).

In recent years, there has been a noteworthy trend towards utilizing pyroligneous extract as a raw material for diverse product generation. Consequently, numerous studies have been conducted not only to employ pyroligneous extract as a fuel but also to explore its applications in medicine, agroindustry, and food (Mohan et al., 2006; Qamar et al., 2023). Despite these advancements, several aspects related to the generation of valuable products from pyroligneous extract require further exploration, particularly concerning the fundamental aspects of generation, development, and optimization of techniques for commercial-scale production. Additionally, it is imperative to stay updated on recent progress in techniques transforming pyroligneous extract into biochemicals and biopolymers (Qamar et al., 2023). Numerous upgrading methods are under investigation, including esterification, catalytic cracking, hot vapor filtration, solvent addition, and supercritical fluid processes (Alhazmi & Loy, 2021; Singh et al., 2023).

The valuable compounds present in the PE can be extracted for various applications, including food flavoring agents, additives for animal feed, antimicrobial agents, preservatives, antioxidants, insect repellants, natural chelating agents, root growth inductors, soil conditioners, pest and disease control agents, and plant development and growth inductors (Loo et al., 2008; Schnitzer et al., 2015; Wang et al., 2012; Wei et al., 2010). The diverse range of applications stems from the different compositions of PE. Table 5 provides insights from studies exploring the use of pyroligneous extracts for various purposes.

In recent years, research has focused on the heterogeneous components of biomass residues, such as cellulose, hemicellulose, and lignin, sparking significant interest as potential feedstock for the production of bioenergy and biochemicals through thermochemical or biochemical approaches, including pyrolysis-related upgrade processes. This provides a sustainable concept for the circular economy (Alhazmi & Loy, 2021; Singh et al., 2023). Among the conversion routes for biomass, pyrolysis has been playing a crucial role in the efficient valorization of biomass (Xue et al., 2022). Exploiting agri-food waste for bioenergy production is a sustainable response to the growing global demand for bioenergy,

**Table 5** Applications of pyroligneous extract (pyroligneous acid or bio-oil) produced from different agricultural wastes

| Biomass | Product | Application | Main results | Reference |
|---|---|---|---|---|
| Fine eulyptus sawdust (*Eucalyptus* spp.) | Pyroligneous acid (PA) | Water repellency and decay resistance of wood | → The formulation containing PA + nanobentonite showed excellent antifungal activity and wood water repellency | Sanson et al. (2022) |
| Cocoa pod shells (*Theobroma cacao* L.) | Pyroligneous acid | Antimicrobial activity | → PA had antimicrobial properties: diameter inhibitory zones ranging from 6 to 6.12 mm for *Candida albicans*, and 6–6.14 mm for *Aspergillus niger* | Desvita & Faisal (2022) |
| Mulberry branches | Pyroligneous acid | Antimicrobial activity | → Antibacterial activity for *Rhizopus* and *Aspergillus* were improved 12.5% and 47.1%, respectively, with the increase of ploidy<br>→ The maximum inhibition rate of PA was 291.7% | Cheng et al. (2022) |
| *Eucommia ulmoides* Oliver branches | Pyroligneous acid | Antimicrobial and antioxidant activity | → PA crude refined by ultra-low freezing and thawing had an excellent inhibition rate (83.33% against *Bacterium prodigiosum*) and a significant effect on scavenging rate (98.72%) for hydroxyl radicals | Xue et al. (2022) |
| Cottonseed | Bio-oil | Biofuel | → Bio-oil had physicochemical properties suitable for engine application<br>→ The best performance and emission characteristics: 5% bio-oil + 95% diesel fuel, compression ratio of 18:1 and 50% of engine load | Rajamohan et al. (2022) |
| Equal mixture of dragon's claw elm (*Ulmus pumila* 'Pendula') and European white elm (*Ulmus laevis* Pall.) | Pyroligneous acid | Herbicide | → In field, pyroligneous acid at 4000 L $ha^{-1}$ provided 100% control of perilla mint, and 82% control of creeping woodsorrel and Carolina geranium.<br>→ Therefore, PA is a great herbicide for broadleaf weed control | Liu et al. (2021a) |
| Appletree prunings (*Malus* × *domestica Borkh.*) | Pyroligneous acid | Herbicide | → At the highest concentration (100%), foliar-applied PA at 2800 L $ha^{-1}$ and soil-applied PA at 110 L $m^{-3}$ can adequately control annual bluegrass<br>→ PA rapidly desiccates annual bluegrass likely due to the high acids content | Liu et al. (2021b) |
| Mushroom substrate waste | Pyroligneous acid | Phytoextraction of Cd/Zn | → The optimal Cd and Zn phytoextraction occurred from the use of 100 times diluted PA, which increased the Cd and Zn extraction by 188% and 164% | Zhou et al. (2022) |
| White pine | Pyroligneous acid | Evaluation of seed priming on germination and seedling growth | → Seed priming with PA can mitigates Al stress and improve tomato seed germination and seedling growth by improving antioxidant defense system | Ofoe et al. (2022) |
| Eucalyptus | Pyroligneous acid | Ecotoxicity evaluation | → PA showed low toxicity for terrestrial plants, but acute toxicity for *D. magna* (crustacean)<br>→ PA did not cause concern regarding its potential to damage genetic material of terrestrial and aquatic organisms (non-genotoxic) | De lima et al. (2019) |

**Table 5** (continued)

| Biomass | Product | Application | Main results | Reference |
|---|---|---|---|---|
| Syagrus coronata's fruit | Pyroligneous acid | Corrosion inhibitor | → PA reached 91% anticorrosive efficiency at 0.6 g $L^{-1}$ | Fernandes et al. (2022) |
| Pine sawdust | Pyroligneous acid | Acid leaching of pine wood for deashing | → The contents of K, Ca, Na, and Mg in pine straw were reduced from 2673, 3285, 287, and 1164 µg/g, respectively to 118, 962, 69, and 105 µg/g, (96%, 71%, 76%, and 90% removal), respectively<br>→ Reduction in ash content from 1.47 wt% to 0.22 wt% | Cen et al. (2023) |
| *Prosopis juliflora* | Bio-oil | As a supplement to phenolic resin in coconut fiber composite reinforcement | → The tensile and flexural properties of the coir fiber reinforced composite laminates admixed with bio-oil and conventional epoxy resin exhibit a higher strength in comparison to that of the neat epoxy resin-based composites | Prabhahar et al. (2022) |
| Wood | Bio-oil and biochar | Production of carbonaceous materials through polymerization | Furfural effectively promoted the cross-polymerisation of bio-oil and biochar, which "bonded" bio-oil and biochar together, forming cross-linking structures. | Hu et al. (2019) |

reducing the reliance on fossil fuels, mitigating greenhouse gas emissions, and decreasing environmental pollution. This approach enriches the energy mix, ensuring safety, and can be considered an ecological fuel (Gholizadeh, Hu and Liuc 2019; Kassim et al., 2022). Bio-oil carries insignificant amounts of sulphur, nitrogen, and ashes, resulting in fewer emissions of pollutants such as $SO_2$, $NO_X$, and particulate matter compared to other liquid fuels like fuel oils and diesel (Dhyani & Bhaskar, 2018). Rajamohan (2022) produced bio-oil from cottonseeds at 339 °C with suitable physicochemical properties for engine use in blends of 5%, 10%, and 20% with diesel fuel, offering good emission and performance characteristics.

Temperature control in pyrolysis is crucial, as temperature variations can either inhibit or activate several compounds in the pyroligneous extracts (Brown and Van Staden, 1997; Mu et al., 2003). At low extraction temperatures (<450 °C), the predominant compounds are alkenes, alkanes, fatty acids, long-chain esters, aliphatic nitriles, and amides (Guedes et al., 2018; Huang et al., 2014). With increasing temperature, alkyl groups of aromatic compounds undergo cleavage through dehydrogenation/aromatization reactions. At higher temperatures, aromatic compounds condense into polycyclic aromatic compounds (Ma et al., 2023; Mahler et al. 2005). In the range of 290 to 370 °C, complete cellulose degradation occurs, resulting in the maximum production rate of acetic acid and volatiles typically around 370 °C. The temperature range between 500 and 900 °C, especially above 700 °C, favors the formation of PAHs (Camargo et al., 2006). Liquid extracts generated above 500 °C raise environmental concerns due to the toxicity of these compounds to aquatic life and their potential carcinogenicity in humans (Mahler et al. 2005).

In the food industry, pyroligneous extract exhibits numerous potential applications, including use as a smoke flavor, antioxidant, and preservative for cheeses, seafoods, meats, and its derivatives through smoking processes (Kim et al., 2012; Loo et al., 2008; Mohan et al., 2006). The antioxidant activity of the extracts can be attributed to polyphenols, which, even in low concentrations, significantly delay or prevent the degradation of subtracts (Loo et al., 2008). Ma et al. (2013) investigated the antioxidant activities, including free-radical

scavenging and the iron-reducing potential of pyroligneous acid (PA) obtained from the pyrolysis of rosemary biomass. Their PAs were diluted in a 2:1 ratio with different organic solvents, and experiments were conducted to evaluate their potential as antioxidants and food preservatives. The highest phenolic concentration was found in the PAs obtained from fresh rosemary leaves, attributed to the higher moisture content of the initial biomass (Ma et al., 2013). It is acknowledged that phenolic compounds contribute to overall food antioxidizing activity and exhibit free-radical scavenging capabilities related to the hydroxyl group (Liu & Yao, 2007).

Pyroligneous extracts exhibit antimicrobial activity against a range of microorganisms, including pathogens, thereby extending the shelf life of both unprocessed and processed foods. The reduction in microbial growth is attributed to the presence of acetic acid and phenols in the PE (Holley & Patel, 2005; Kim et al., 2012; Thurette et al., 1998). Even at low concentration, phenolic compounds and acetic acid demonstrate the ability to reduce microbial activity under conditions that align with good practices in food production (Thurette et al., 1998). In some Asian countries, PE is considered an antiseptic, anti-inflammatory, antioxidant, and antiallergic agent. Leveraging its detoxifying properties, the product, when applied to the human body, is believed to enhance blood circulation and promote a detoxifying effect (Kim et al., 2012; Loo et al., 2008). The antimicrobial activity of PE, primarily attributed to phenolic compounds, also contributes to wound healing (Kim et al., 2012). The phenolic content plays a direct role in antioxidizing activity, exhibiting inhibitory effects on human mutagenesis and carcinogenesis (Loo et al., 2008).

Infectious diseases caused by pathogenic microorganisms remain significant in developing countries, posing threats to both humans and plants. Many antifungal agents contain substances that are challenging to decompose, potentially contributing to environmental pollution (Desvita & Faisal, 2022). Therefore, the exploration of new substances with potential antimicrobial activity derived from natural products becomes crucial. Desvita & Faisal (2022) produced a pyroligneous acid (PA) from cocoa shells in the temperature range of 300–380 °C. Their results demonstrate that the PA inhibited the growth of two microbial pathogens, *Candida albicans* and *Aspergillus niger*, with higher PA concentrations leading to a larger inhibition zone. The inhibition can be attributed to specific components of the PA, such as phenol and acetic acid, which interfere with the cell membranes of fungi, increasing the permeability of the cell wall and, in some cases, leading to a loss of cell contents. Given these findings, PA is a potential antimicrobial agent, possibly replacing conventional antibiotics (Desvita & Faisal, 2022).

According to Xue et al. (2022), the phenolic and ketone groups, mainly phenol, eugenol, and guaiacol, have a strong antioxidant and free radical scavenging capacity. Results demonstrated inhibition rates higher than 73% against *Escherichia coli, Staphylococcus aureus, Enterobacter aerogenes, Bacillus subtilis, Bacterium prodigiosum* and *Proteusbacillus vulgaris* using pyroligneous acid, indicating that this product can be applied as a natural antibiotic (Xue et al., 2022). The PA produced by Cheng et al. (2022) showed different degrees of inhibition for five bacteria, which was related to the percentage of phenolic compounds, acetic acid and ketones in each extract. PA from mulberry branches has an appreciable potential as a natural antibacterial agent that can be explored in medicine, agriculture, cosmetic and food industry (Cheng et al., 2022). In addition, PA has different antifungal activity against pathogenic fungi, which can be attributed to cellulose and lignin content of the biomass, as well as guaiacyl nuclei. The inhibitory effects can be attributed to the formation

of compounds with antimicrobial effects generated by oxidation, dehydration, condensation, and rearrangement reactions. The PA can suppress the bacterial and pathogenic fungi growth due to the toxicity of formaldehyde, acetic acid, and methanol present in the liquids (Yan et al., 2020). In this context, biocides with lower environmental and human health impacts are increasingly demanded by society.

In agriculture, pyroligneous extract shows great potential both in its raw form and as an ingredient in novel agricultural formulation (Gajić & Koch, 2012). Various studies have highlighted the efficiency of PE as a potential agricultural product. PE can be utilized for pest and disease control, enhancing plant development and growth, acting as an insect repellant, weed growth inhibitor, natural chelating agent, organic fertilizer, promoting root growth, serving as a soil conditioner, and acting as an additive for animal feed (Kim et al., 2012; Loo et al., 2008; Schnitzer et al., 2015; Wang et al., 2012; Wei et al., 2010; Zanetti et al., 2004). Pyroligneous extracts with a high concentration of organic acids and phenolic compounds exhibit significant antimicrobial activity and antioxidizing activities, respectively (Ma et al. 2011; Wu et al., 2015). Additionally, acetic acid and phenolic compounds demonstrate anti-germinating, antifungal, and anti-termite properties (Baimark & Niamsa, 2009). Under certain production conditions, pyroligneous acid exhibit low toxicity and contains molecules with chelating properties. These substances can enhance the efficiency of phytosanitary products and the absorption of nutrients from fertilizers when used in leaf pulverization, leading to benefits for plant growth and health (Zanetti et al., 2004).

As a soil conditioner, PE can enhance the physical, chemical, and, most importantly, the biological properties of the soil, increasing the population of beneficial microorganisms and facilitating nutrient uptake by plants (Schnitzer et al., 2015). A prevalent issue affecting 50% of global agricultural land is the presence of acid soils (Ofoe et al., 2022). These soils are characterized by the abundance of triple-charged aluminum ions ($Al^{3+}$), leading to various toxicity syndromes in plants. The risk associated with such soil stress is exacerbated by climate changes and the intensive use of synthetic fertilizers (Bojórquez-Quintal et al., 2017). In this context, studying the incorporation of PAs into the soil is crucial. PA can be rich in bioactive compounds, serving as an essential tool to support plant growth and enhance plant tolerance to environmental stress in acid soils (Ofoe et al., 2022). Ofoe et al. (2022) investigated the effect of White pine PAs in tomato seed growth. They observed a significant increase in germination rate of seeds, sapling vigor, total plant height, hypocotyls and root surface areas, total root volume, and saplings mass.

Pyroligneous acid also demonstrates potential as an herbicide (Liu et al., 2021a, b). In field experiments, PA exhibited varying degrees of control over perilla mint, creeping sorrel, and Carolina geranium, suggesting its utility as a non-synthetic herbicide for managing broadleaf weeds (Liu et al., 2021a). Observations indicate that PA is more effective against annual broadleaf weeds in their early stages of development, showing significantly less effective on both annual and perennial weeds in mature stages. The herbicidal capacity of PA can be attributed mainly to acids, particularly acetic acid, identified as the primary water-soluble organic active compound in the liquid (Brainard et al. 2013; Liu et al., 2021a). Aguirre et al. (2020) suggested that other organic compounds detected in PA, such as aldehydes, ketones, furans, and phenols, may also contribute to its herbicidal activity. Regions with high agricultural production are promising places for the development of such new products derived from pyroligneous extracts, enabling increased sustainability with the use of pyrolysis of agro-waste on a large scale. Based on all its potential applications, the pyroligneous

extract is a promising product for various industrial purposes, including agricultural, energy, medicinal and food uses (Fig. 3).

Although it has several potential uses (Fig. 3), pyroligneous extract can present non-advantageous characteristics that prevent its direct use in advanced sectors, requiring other steps to remove unwanted compounds or enrich valuable content (Chai et al., 2024). In order to overcome its limitations, advanced pyrolysis methods have been studied to improve the properties, characteristics, and yield of the pyrolysis products, while enhancing operational parameters. These methods mainly include catalytic pyrolysis, co-pyrolysis, hydropyrolysis, and microwave pyrolysis (Cai et al., 2024). Fast pyrolysis, which presents bio-oil as the main product generated (70–80% by weight), is still in the research and development phase as certain challenges need to be overcome in the future, including high production costs due to the high operating temperature demands, short residence time, and subsequent deoxygenation, instability, bio-oil quality, among other factors (Alhazmi & Loy, 2021; Singh et al., 2023).

## 5 Research needs and future directions

Pyrolysis stands out as an innovative technology for the sustainable recovery of resources from agricultural wastes. Its application not only helps prevent environmental pollution but also enables the acquisition of economic and environmental benefits through the conversion of agricultural waste into value-added products (Silos-Llamas et al., 2023). The pyrolysis of agro-wastes presents opportunities for the production of solids, liquids, and gases, each with distinct properties and compositions suitable for various applications. However, the inherent variability in the properties of its products poses challenges for pyrolysis in terms of technological updates, applications, and commercialization.

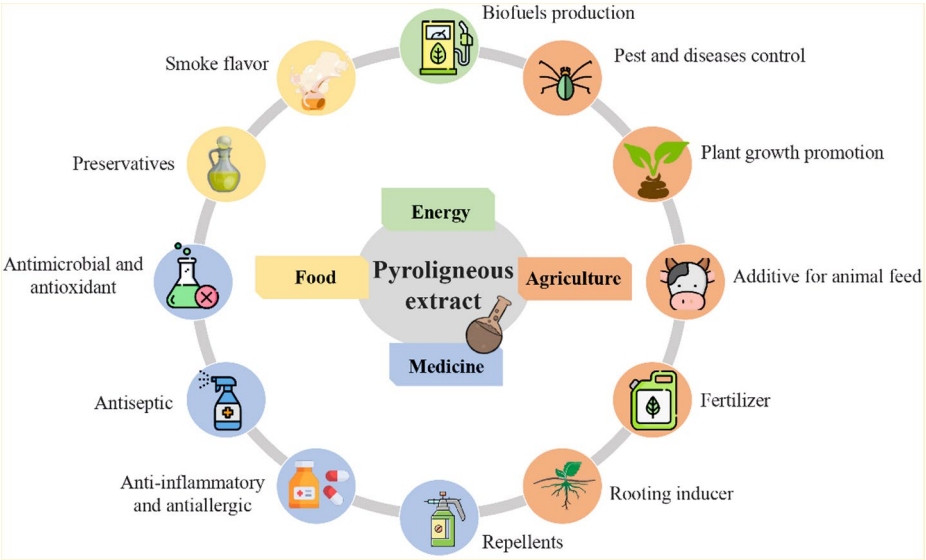



**Fig. 3** Uses of pyroligneous extracts in agricultural, food, energy and medicinal applications

The field of research on the pyrolysis of agro-wastes remains open and dynamic, considering the diversity of agricultural residues, equipment, operational conditions, and resulting products. The complexity of pyrolysis studies is heightened by challenges in reproducing and comparing results, coupled with the extensive range of techniques required for characterizing pyrolysis products, particularly liquids. Anticipating numerous environmental benefits and predicting low production costs, it is expected that an increasing number of studies will delve into the generation of pyrolysis products with specific characteristics. Additionally, there is a growing need for the further development of applications to facilitate their integration into industrial scales, accompanied by the formulation of specific regulations.

Biochar has garnered substantial interest for its diverse applications, especially in environmental, energy, and agricultural sectors. Notably, the majority of studies in these applications are conducted at the laboratory scale. Consequently, the actual utilization of biochar remains an emerging area with numerous research needs to unlock its full potential across various fields. Future research endeavors should focus on exploring methods involving chemical or physical activation, flash, vacuum, and microwave pyrolysis to enhance biochar characteristics and the overall pyrolysis process. Furthermore, there is a pressing need for a more in-depth exploration of characterization techniques to accurately assess the diverse physicochemical and surface properties of biochar. Ultimately, these advancements are crucial for facilitating the widespread and effective incorporation of biochar on a larger scale across different sectors.

While pyroligneous extract holds significant potential for energy applications, its quality falls short when compared to fossil fuels across various parameters, including high oxygen content, elevated moisture content, low calorific value, and increased viscosity. Consequently, its direct utilization is not presently recommended. Future studies, particularly in the field of pyroligneous extract application, should aim to expand its uses into agriculture, medicine, and the food industry. Additionally, there should be a heightened focus on exploring value-added products derived from agro-waste pyrolysis. Emphasizing commercialization and stakeholder engagement is crucial to stimulate interest in pyroligneous extract utilization. Numerous upgrading technologies have been developed to enhance its quality, as outlined earlier. Furthermore, comprehensive experimental research is essential to evaluate the ideal pyrolysis conditions, considering factors such as biomass origin, production methods, economic aspects, and target markets for the sustainable application and commercialization of pyroligneous extract and other products derived from pyrolysis. Moreover, it is crucial to establish a continuous supply chain for the pyrolysis of agricultural wastes. This step is essential to strengthen and commercialize the utilization of pyrolysis products in real-world conditions, contributing to the attainment of sustainable development goals through their numerous promising applications.

## 6 Concluding remarks

Pyrolysis is a promising technology for the thermochemical conversion of agricultural wastes due to its cost-effectiveness, simplicity in operation, and the range of products obtained (solid, liquid and gases). Researchers worldwide have been exploring ways to enhance the viability of pyrolysis, either as a solution for waste management or for the development of value-added products for various purposes.

Product yields and applications vary according to process conditions and parameters, such as temperature, heating rate and residence time, type of reactor, pyrolyzed agro-waste, among others. Pyrolysis products have numerous potential applications in the environmental, pharmaceutical, agricultural, and food sectors. Biochar, the most extensively studied pyrolysis product, has diverse applications, including (i) removal of contaminants from aquatic environments, soils, and air; (ii) carbon sequestration; (iii) bioenergy production; (iv) soil amendment; (v) as an additive for animal feed and composting; among others. The non-condensable gases generated during pyrolysis are commonly utilized for heat or electricity generation. The pyroligneous extract, composed of pyroligneous acid and bio-oil, has garnered increased attention in recent years, with diverse uses being proposed and tested. It has been recognized as a promising product with applications in energy (specifically in biofuel production), agriculture (for promoting plant growth, controlling pests and diseases, and as an additive for animal feed), medicine (with potential antimicrobial, antioxidant, anti-inflammatory, and antiallergic properties) and food industry (as a preservative and smoke flavor).

Therefore, pyrolysis is a promising technology for agricultural waste valorization in the circular economy context. The utilization of pyrolysis products for the purposes discussed in this work is capable of promoting several Sustainable Development Goals (SDGs), especially SDG 2 (Zero hunger), SDG 3 (Good health and well-being), SDG 6 (Clean water and sanitation), SDG 7 (Affordable and clean energy), SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable cities and communities), SDG 12 (Responsible consumption and production), SDG 13 (Climate action), SDG 14 (Life below water) and SDG 15 (Life on Land).

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**Data availability** Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

## Declarations

**Competing interest** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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